

Customer Shopping Behaviour

Data Analysis Case Study

Retail trends, customer loyalty, and product performance

Dataset: 3,900 Customers | 18 Variables

Tools: PostgreSQL • Power BI

Problem Statement

What are the key drivers of customer spending behaviour, and how can the business use purchase patterns, loyalty segmentation, and product performance to improve revenue and customer retention?

This case study analyses a retail dataset of 3,900 customers across 18 variables — covering demographics, purchase history, product preferences, shipping habits, and subscription status. The goal is to extract actionable insights around revenue performance, customer loyalty, product trends, and the impact of discounts and subscriptions on spending.

Data Overview

About the Dataset

The dataset contains 3,900 customer transaction records from a US retail business. Each record represents a single customer and their most recent purchase, along with historical context such as previous purchases and subscription status.

Data Dictionary

Column	Description	Scale / Values
Customer ID	Unique identifier for each customer	Numeric
Age	Age of the customer	18 — 70
Gender	Gender of the customer	Male / Female
Item Purchased	Product bought	25 unique items
Category	Product category	Clothing / Accessories / Footwear / Outerwear
Purchase Amount (USD)	Transaction value	\$20 — \$100
Location	US state of purchase	Multiple states
Size	Product size	XS / S / M / L / XL
Color	Product colour	Multiple colours
Season	Season of purchase	Spring / Summer / Fall / Winter
Review Rating	Customer rating	2.5 — 5.0
Subscription Status	Newsletter subscriber	Yes / No

Column	Description	Scale / Values
Shipping Type	Shipping method	6 types
Discount Applied	Discount used	Yes / No
Promo Code Used	Promo code applied	Yes / No
Previous Purchases	Number of past purchases	1 — 50
Payment Method	Method of payment	Multiple methods
Frequency of Purchases	How often they shop	Weekly / Fortnightly / etc.

Summary Statistics

A snapshot of the overall dataset before diving into the analysis:

Metric	Value
Total Customers	3,900
Average Age	44.1 years
Male / Female Split	68% / 32%
Average Purchase Amount	\$59.76
Average Review Rating	3.75 / 5.0
Average Previous Purchases	25.4
Subscribed Customers	27% (1,053)
Discount Applied	43% of transactions
Top Category by Revenue	Clothing (\$104,264)
Most Purchased Product	Blouse / Jewelry / Pants (171 each)
Least Purchased Product	Jeans (124 orders)
Highest Rated Product	Gloves (3.86 / 5.0)

The average customer spends \$59.76 per transaction, rates their purchase 3.75 out of 5, and has made 25 previous purchases. 73% are not subscribed to the newsletter, and Clothing is by far the most purchased category at 44.7% of total revenue.

Section 1 — Revenue & Demographics

Q1. What is the total revenue generated by male vs female customers?

Compares total revenue contribution by gender.

```
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender;
```

Results:

Gender	Total Revenue	% of Total
Male	\$157,890	67.7%
Female	\$75,191	32.3%

Finding: Male customers generate more than double the revenue of female customers (\$157,890 vs \$75,191). This likely reflects the higher proportion of male customers in the dataset (2,652 male vs 1,248 female) rather than higher individual spend.

Q10. What is the revenue contribution of each age group?

Breaks revenue down by four age bands to identify the most valuable customer demographics.

```
SELECT age_group, SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;
```

Results:

Age Group	Total Revenue
Senior (55+)	\$69,590
Middle-aged (40–54)	\$67,188
Adult (25–39)	\$67,045

Age Group	Total Revenue
Young Adult (18–24)	\$29,258

Finding: Revenue is remarkably even across Senior, Middle-aged, and Adult groups ($\pm 1\%$ of each other), while Young Adults contribute significantly less. This suggests the business’s product range resonates most with customers aged 25 and above.

Section 2 — Product & Category Performance

Q3. Which are the top 5 products with the highest average review ratings?

```
SELECT item_purchased,  
       ROUND(AVG(review_rating::numeric), 2) AS avg_review_rating  
FROM customer  
GROUP BY item_purchased  
ORDER BY AVG(review_rating) DESC  
LIMIT 5;
```

Results:

Product	Avg Review Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.79

Finding: Gloves, Sandals, and Boots are the highest-rated products. Notably, two of the top five (Sandals, Boots) are Footwear items, suggesting that customers are particularly satisfied with footwear despite it being the third-largest revenue category.

Q12. What are the top 5 most purchased products?

```
SELECT item_purchased, COUNT(customer_id) AS total_orders  
FROM customer  
GROUP BY item_purchased  
ORDER BY total_orders DESC  
LIMIT 5;
```

Results:

Product	Total Orders
Blouse	171
Jewelry	171
Pants	171
Shirt	169
Dress	166

Finding: The top 5 most purchased products are extremely tightly clustered (166–171 orders each), suggesting even demand distribution across the core clothing and accessories range. No single product dominates.

Q13. What are the bottom 5 least purchased products?

```
SELECT item_purchased, COUNT(customer_id) AS total_orders
FROM customer
GROUP BY item_purchased
ORDER BY total_orders ASC
LIMIT 5;
```

Results:

Product	Total Orders
Sneakers	145
Boots	144
Backpack	143
Gloves	140
Jeans	124

Finding: Jeans is the least purchased product (124 orders), and notably Gloves and Boots appear in both the least purchased and highest-rated lists. High satisfaction does not guarantee high volume — these are niche but well-loved products.

Q8. What are the top 3 most purchased products within each category?

```

WITH item_counts AS (
    SELECT category, item_purchased,
           COUNT(customer_id) AS total_orders,
           ROW_NUMBER() OVER(PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS
item_rank
    FROM customer
    GROUP BY category, item_purchased
)
SELECT item_rank, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;

```

Results:

Rank	Product	Category	Total Orders
1	Blouse	Clothing	171
2	Shirt	Clothing	169
3	Dress	Clothing	166
1	Jewelry	Accessories	171
2	Handbag	Accessories	165
3	Sunglasses	Accessories	163
1	Sandals	Footwear	158
2	Sneakers	Footwear	145
3	Boots	Footwear	144
1	Coat	Outerwear	89
2	Jacket	Outerwear	85
3	Hoodie	Outerwear	83

Finding: Blouse and Jewelry are tied as the most purchased items overall (171 each). Within Footwear, Sandals lead despite the category being third in total revenue. Outerwear has the lowest per-product volumes, consistent with it being the smallest revenue category.

Q11. What is the top selling product in each category?

```

WITH ranked_items AS (
    SELECT category, item_purchased,
           COUNT(customer_id) AS total_orders,

```



```

        ROW_NUMBER() OVER(PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS
rank
    FROM customer
    GROUP BY category, item_purchased
)
SELECT category, item_purchased, total_orders
FROM ranked_items
WHERE rank = 1
ORDER BY total_orders DESC
LIMIT 5;

```

Results:

Category	Top Product	Total Orders
Clothing	Blouse	171
Accessories	Jewelry	171
Footwear	Sandals	158
Outerwear	Coat	89

Section 3 — Subscriptions & Customer Loyalty

Q5. Do subscribed customers spend more?

```
SELECT subscription_status,  
       COUNT(customer_id) AS total_customers,  
       ROUND(AVG(purchase_amount), 2) AS avg_spend,  
       ROUND(SUM(purchase_amount), 2) AS total_revenue  
FROM customer  
GROUP BY subscription_status  
ORDER BY total_revenue, avg_spend DESC;
```

Results:

Subscription	Customers	Avg Spend	Total Revenue
No	2,847	\$59.87	\$170,436
Yes	1,053	\$59.49	\$62,645

Finding: Subscribed customers spend virtually the same per transaction as non-subscribers (\$59.49 vs \$59.87 — a difference of just \$0.38). The lower total revenue for subscribers simply reflects that only 27% of customers are subscribed. Subscription status alone does not predict higher individual spend.

Q7. Customer segmentation by purchase history

Segments customers into New, Returning, and Loyal based on previous purchases. New = 1 purchase, Returning = 2–10, Loyal = 11+.

```
WITH customer_type AS (  
  SELECT customer_id, previous_purchases,  
         CASE  
           WHEN previous_purchases = 1 THEN 'New'  
           WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'  
           ELSE 'Loyal'  
         END AS customer_segment  
  FROM customer  
)  
SELECT customer_segment, COUNT(*) AS "Number of Customers"  
FROM customer_type  
GROUP BY customer_segment;
```

Results:

Segment	Number of Customers	% of Total
Loyal	3,116	79.9%
Returning	701	18.0%
New	83	2.1%

Finding: 79.9% of customers are classified as Loyal (11+ previous purchases), which is an exceptionally high retention rate. Only 2.1% are New customers, suggesting this business retains its customer base effectively but may be struggling with new customer acquisition.

Q9. Are repeat buyers more likely to subscribe?

Looks at subscription rates among customers with more than 5 previous purchases.

```
SELECT subscription_status, COUNT(customer_id) AS repeat_buyer
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

Results:

Subscription Status	Repeat Buyers
No	2,518 (72.4%)
Yes	958 (27.6%)

Finding: Even among repeat buyers (5+ purchases), 72.4% are not subscribed. This is a significant gap — loyal customers who have already demonstrated commitment to the brand are not being converted to subscribers. A targeted retention campaign for this group could meaningfully grow the subscriber base.

Section 4 — Discounts, Shipping & Spending Patterns

Q2. Which customers used a discount but still spent above average?

Identifies customers who applied a discount yet still exceeded the average purchase amount of \$59.76.

```
SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied = 'Yes'
AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);
```

Finding: A meaningful segment of discount users still spend above average, suggesting discounts are not eroding transaction value across the board. These customers represent high-value buyers who respond to promotions without being price-dependent.

Q6. Which 5 products have the highest discount usage rate?

```
SELECT item_purchased,
       ROUND(SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END) / COUNT(*) *
             100, 2) AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

Results:

Product	Discount Rate
Blouse	50.29%
Pants	49.71%
Jewelry	49.71%
Hat	49.37%
Dress	49.40%

Finding: The top discounted products mirror the top purchased products almost exactly — Blouse, Pants, Jewelry, and Dress all appear in both lists. Discounts appear to be driving volume for already popular items rather than shifting demand to underperforming products.

Q4. Does shipping type affect average spend?

Compares average purchase amounts between Standard and Express shipping.

```
SELECT shipping_type,  
       ROUND(AVG(purchase_amount), 2) AS avg_spend  
FROM customer  
WHERE shipping_type IN ('Standard', 'Express')  
GROUP BY shipping_type;
```

Results:

Shipping Type	Avg Purchase Amount
Express	\$60.48
Standard	\$58.46

Finding: Express shipping customers spend on average \$2.02 more per transaction than Standard customers. While modest, this pattern suggests customers willing to pay for faster delivery may also be more willing to spend more on their purchases overall.

Power BI Dashboard

The following three dashboard pages visualise the key findings across sales performance, customer loyalty, and product analysis.

Page 1 — Sales Overview

A high-level overview of total customers, average spend, review ratings, revenue by category, subscription breakdown, and revenue by age group.

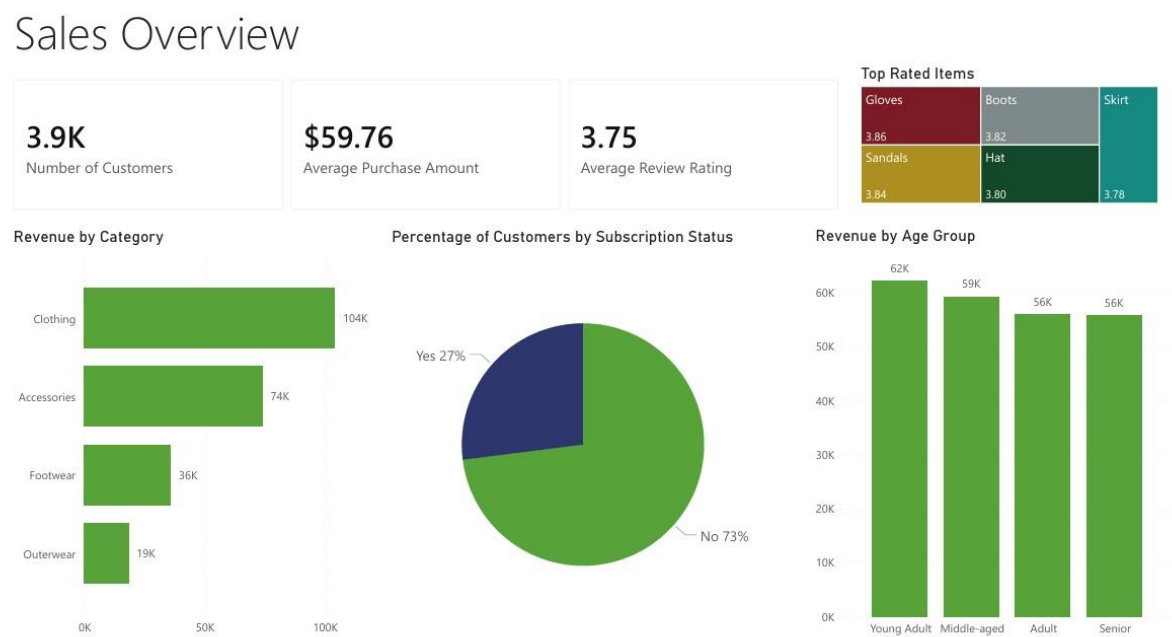


Figure 1 — Sales Overview Dashboard

Finding: Clothing dominates revenue at \$104K (44.7% of total). Revenue is evenly distributed across age groups aged 25+, and 73% of customers are non-subscribers.

Page 2 — Customer Loyalty

Breaks down customer loyalty segmentation, subscription rates among repeat buyers, and spend comparison between subscribers and non-subscribers.

Customer Loyalty

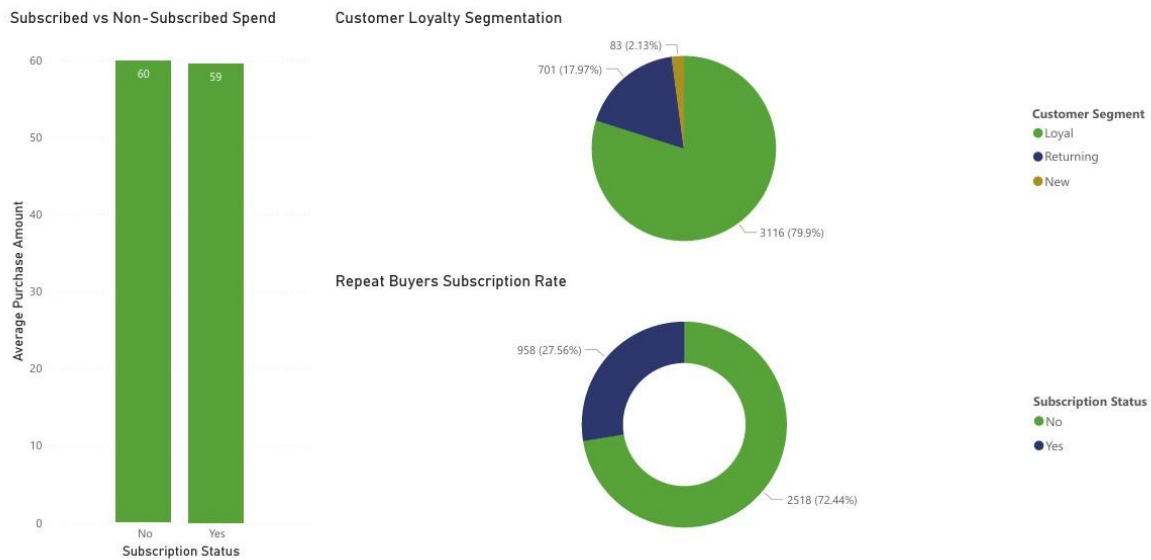


Figure 2 — Customer Loyalty Dashboard

Finding: 79.9% of customers are Loyal (3,116 customers). Despite high loyalty, 72.4% of repeat buyers have not subscribed — a significant conversion opportunity.

Page 3 — Products and Sales Performance

Shows the most and least popular products by purchase volume, and revenue contribution by category.

Products and Sales Performance

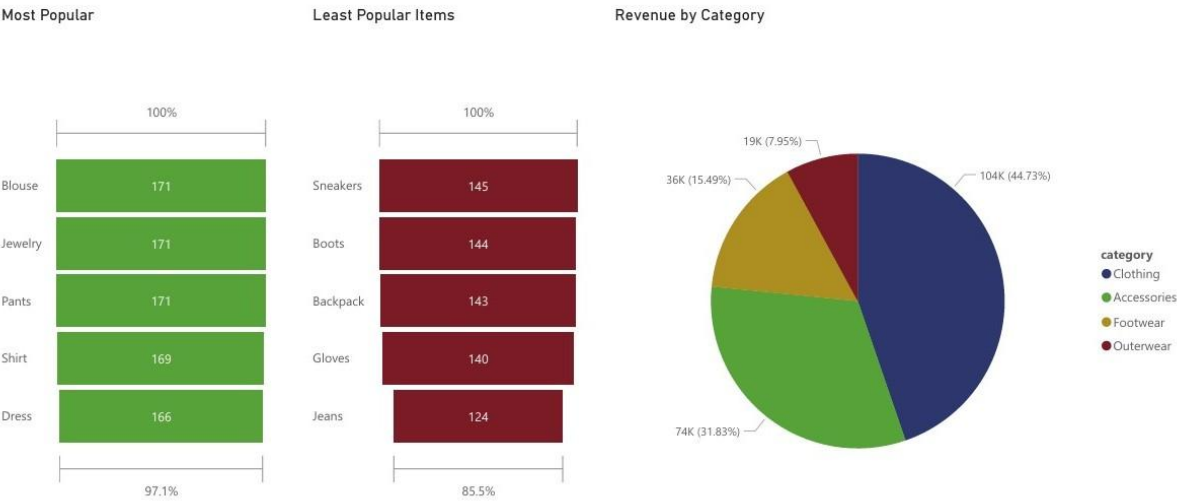


Figure 3 — Products and Sales Performance Dashboard

Finding: Blouse, Jewelry, and Pants are the top sellers at 171 orders each. Jeans is the weakest performer at 124 orders. Clothing accounts for 44.7% of total revenue.

Conclusion & Recommendations

Key Findings

- Male customers account for 67.7% of total revenue, primarily reflecting their larger share of the customer base rather than higher individual spend.
- Revenue is evenly spread across customers aged 25–70, with Young Adults (18–24) contributing notably less.
- 79.9% of customers are classified as Loyal, indicating strong retention — but only 2.1% are New customers, pointing to a new acquisition gap.
- Subscribed and non-subscribed customers spend virtually the same per transaction (\$59.49 vs \$59.87) — subscription does not drive higher individual spend.
- 72.4% of repeat buyers are not subscribed, representing a clear conversion opportunity.
- Discounts are concentrated on already top-performing products, rather than boosting weaker ones.
- Express shipping customers spend slightly more on average than Standard customers.
- Gloves and Boots are among the least purchased products but have the highest customer satisfaction ratings.

Recommendations

- **Target repeat buyers with a subscription conversion campaign** — with 2,518 loyal non-subscribers, even a 20% conversion rate would nearly double the subscriber base.
- **Redirect discount strategy** — discounts are currently concentrated on top sellers. Applying them to underperforming products like Jeans (124 orders) could help balance the product portfolio.
- **Invest in young adult acquisition** — the 18–24 age group contributes significantly less revenue. Targeted campaigns or product lines aimed at this demographic could unlock a new revenue stream.
- **Promote high-rated low-volume products** — Gloves and Boots have top satisfaction scores but low purchase volumes. Increased visibility or bundling could convert satisfaction into sales.

Closing Statement

This business has an exceptionally loyal customer base but faces two structural challenges: low new customer acquisition (only 2.1% of customers are new) and a large pool of loyal non-subscribers who have not been converted despite high engagement. Addressing these two gaps — through targeted acquisition and a subscription conversion campaign — represents the clearest path to sustainable revenue growth.